



THE KING'S SCHOOL

2003
Higher School Certificate
Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

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Attempt Questions 1-8
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Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Find

(i) $\int \frac{1+x+x^2}{1+x^2} dx$ 2

(ii) $\int \frac{x^2}{1+x^2} dx$ 2

(b) Use integration by parts to evaluate

$\int_0^1 2x \tan^{-1} x \, dx$ 3

(c) Find $\int_0^1 \frac{x-3}{(x^2+1)(3x+1)} dx$, giving your answer in simplest exact form. 4

(d) $u_n = \int_0^1 \frac{x^n}{1+x^2} dx, n \geq 0$

(i) Show that $u_{n+2} + u_n = \frac{1}{n+1}$ 2

(ii) Hence, evaluate $\int_0^1 \frac{x^3}{1+x^2} dx$ 2

End of Question 1

Marks

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) $u = 2+ai, v = a+2i$, where a is a real number.

Find in the form $x+iy$,

(i) uv 2

(ii) $(uv)^{-1}$ 1

(b) (i) Express $z = -2\sqrt{3} + 2i$ in modulus-argument form 2

(ii) Hence, find z^3 in the form $x+iy$ 2

(c) Sketch the region in the complex plane where

$|z-i| \leq |z+1|$ 3

(d) Consider the equation $(a+ib)^2 = 1+2i$, a, b real

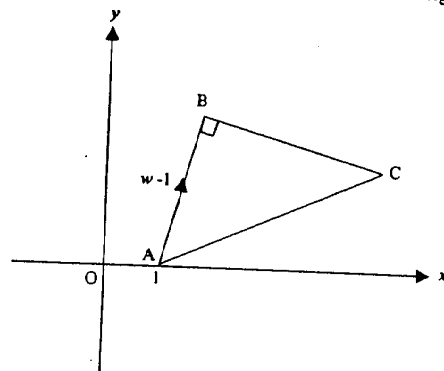
(i) Show that $a^2 + b^2 = \sqrt{1^2 + 2^2}$ 1

(ii) Hence, or otherwise, find the value of a^2 2

Question 2 continues on next page

Question 2 (continued)

- (e) In the complex plane, A is the point (1,0) and the complex number $\overline{AB}$ is $w-1$. $\triangle ABC$ is isosceles and right-angled at B. O is the origin.



Find, in terms of w , the complex numbers

(i) $\overline{CB}$

(ii) $\overline{OC}$

End of Question 2

Marks

1

1

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Sketch on the same axes the graphs of

$$y = |x-1| \text{ and } y = 2x - x^2$$

2

- (ii) Use (i) to show on separate diagrams, the graphs of

(α) $y = \frac{|x-1|}{2x-x^2}$, showing any asymptotes

3

(β) $y = \frac{2x-x^2}{|x-1|}$, showing any asymptotes

3

- (b) Consider the function $f(x) = \tan^{-1} x - \frac{x}{1+x^2}$

- (i) Show that f is an odd function.

1

- (ii) Find $f'(x)$

2

- (iii) Show that $f(x) > 0$ if $x > 0$

2

- (iv) Sketch the graph of $y = \tan^{-1} x - \frac{x}{1+x^2}$

2

End of Question 3

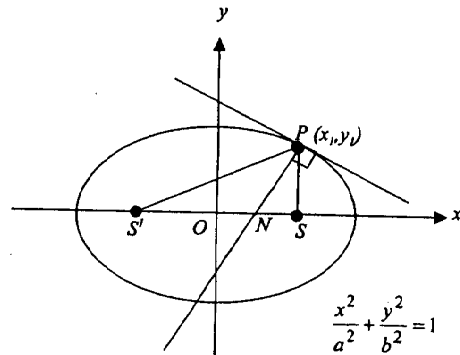
Marks

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the gradient of the tangent to the curve $x^3 + y^2 + xy = 0$ at the point $(-2, 4)$ 3

- (b) $P(x_1, y_1)$ is a point in the first quadrant on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$

S and S' are the foci of the ellipse. O is the origin.



- (i) Show that the equation of the normal at $P(x_1, y_1)$ is $a^2 y_1 (x - x_1) = b^2 x_1 (y - y_1)$ 2

- (ii) The normal at P meets the major axis at N .

Prove that the x coordinate at N is $e^2 x_1$, where e is the eccentricity of the ellipse. 2

- (iii) Deduce that N lies between O and S . 2

- (iv) Show that $NS = eSP$ and $NS' = eS'P$ 3

- (v) Using the sine rule in $\triangle PSN$ and $\triangle PS'N$, or otherwise, prove that PN bisects $\angle SPS'$ 3

End of Question 4

Marks

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) Four married couples are to be seated at a circular table.

- (i) How many arrangements are possible if the men and women are to be separated? 2

- (ii) For the arrangements in (i), find the probability that no woman is sitting next to her husband. 2

- (b) The equation $x^3 + ax^2 + bx + c = 0$ has one root the sum of the other two roots.

Prove that $a^3 - 4ab + 8c = 0$ 4

- (c) (i) By considering the circle $x^2 + y^2 = a^2$, or otherwise, find

$$\int_0^a \sqrt{a^2 - x^2} dx$$

2

- (ii) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b > 0$, is revolved about the line $y = a$.

By considering slices perpendicular to the line $y = a$, find the volume of the solid of revolution generated. 5

End of Question 5

Marks

Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle of mass m kg falls vertically from rest from point O in a medium whose resistance is mkv , where k is a positive constant and v is its velocity in m/s. After t seconds the particle has fallen x metres.

g m/s² is the acceleration due to gravity.

(i) Show that $\frac{dv}{dt} = g - kv$

1

- (ii) Find the terminal velocity, V m/s, of the particle.

1

(iii) Use integration to prove that $v = \frac{g}{k} (1 - e^{-kt})$

3

- (iv) Find the distance the particle has fallen when its velocity is one half of its terminal velocity.

4

- (b) α , β are the two complex roots of the equation $x^3 + 5x + 1 = 0$

- (i) Explain why α , β are complex conjugates.

1

(ii) Show that the real root is $\frac{-1}{|\alpha|^2}$

2

- (iii) Show that $\alpha\beta$ is a root of the equation $x^3 - 5x^2 - 1 = 0$

3

End of Question 6

Marks

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) By mathematical induction it is easy to show that

$$1^2 - 2^2 + 3^2 - \dots - (2n)^2 = -n(2n+1)$$

If, further, it is known that

$$1^2 + 2^2 + 3^2 + \dots + (2n)^2 = \frac{n}{3}(2n+1)(4n+1),$$

deduce that

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n}{6}(n+1)(2n+1)$$

(Do not use induction)

3

- (b) (i) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x}$$

3

- (ii) Let $F(x)$ be a primitive function of $f(x)$.

Using this, or otherwise, show that

$$\int_0^{2a} f(x) dx = \int_0^a f(x) + f(2a-x) dx$$

2

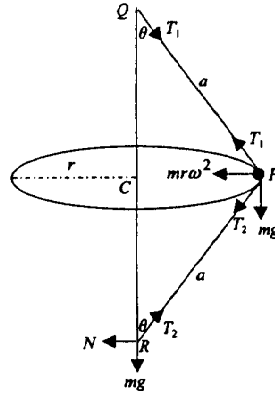
- (iii) Deduce $\int_0^{\frac{\pi}{2}} \frac{x}{1 + \sin x} dx$

3

Question 7 continues next page

Question 7 (continued)

- (c) A mass m at P is freely joined to two equal light rods PQ and PR of length a . The end Q of PQ is pivoted to a fixed point Q and the end R of PR is freely joined to a ring of mass m which slides on a smooth vertical pole. If P rotates in a horizontal circle with uniform angular velocity ω , show the angle of inclination of the rods PQ and PR to the vertical is $\tan^{-1}\left(\frac{r\omega^2}{3g}\right)$. T_1, T_2 are tensions in the rods, N is the normal reaction of QR on the ring R .



End of Question 7

4

Question 8 (15 marks) Use a SEPARATE writing booklet.

- (a) The roots of $z^n = 1$, n a positive integer, are

$$z_k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}, \quad k = 1, 2, \dots, n$$

- (i) Show that $z_k^n = z_1^{kp}$, p a positive integer

2

- (ii) If z_k is such that $z_k, z_k^2, z_k^3, \dots, z_k^n$ generates all the roots of $z^n = 1$, then z_k is called a primitive root of $z^n = 1$

- (α) Show that z_1 is a primitive root of $z^n = 1$

1

- (β) Show that z_5 is a primitive root of $z^6 = 1$

2

- (γ) Suppose the highest common factor of n and k is h , ie, $n = ph$ and $k = qh$, p, q integers.

Show that for z_k to be a primitive root of $z^n = 1$, then $h = 1$

2

- (b) (i) Show that $\sum_{k=0}^{n-1} (1-x)^k = \frac{1-(1-x)^n}{x}$, $x \neq 0$

2

- (ii) Deduce that $\sum_{k=0}^{n-1} (1-x)^k = \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} x^{k-1}$

2

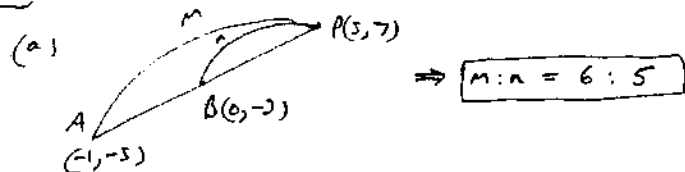
- (iii) Explain or show why $\int \sum_{k=0}^{n-1} (1-x)^k dx = \sum_{k=0}^{n-1} \int (1-x)^k dx$

1

- (iv) Deduce that $\sum_{k=1}^n (-1)^{k-1} \binom{n}{k} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$

3

Qn 1



(b) $\frac{d}{dx} \tan^{-1}(1+x^2) = \frac{1}{1+(1+x^2)^2} \times 2x = \frac{2x}{1+(1+x^2)^2}$

(c) $\binom{8}{5}$ or, of course, $\binom{8}{3} = 56$

(d) gradients of lines are 2 and -3
 $\therefore \tan \theta = \left| \frac{2 - (-3)}{1 + 2(-3)} \right| = \frac{5}{5} = 1$
 $\therefore$ acute angle is 45°

(e) $P(-1) = 0 \Rightarrow (-1)^{2n+1} - (-1)^{2n} + b = 0$
 $\therefore -1 - 1 + b = 0 \therefore b = 2$

(f) $\sum_{n=1}^9 \left(\frac{1}{n} - \frac{1}{n+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{9} - \frac{1}{10} \right)$
 $= 1 - \frac{1}{10}$
 $= \frac{9}{10}$

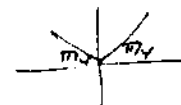
Qn 2

(a) $f'(x) = 5 - 4 \cos 4x$
 $\geq 5 - 4(1)$ since $-1 \leq \cos 4x \leq 1$
 $\Rightarrow f'(x) \geq 1 > 0 \quad \forall x$
 $\therefore f(x)$ increases $\forall x$

(b) (i) $R \sin(x-\alpha) = R \cos \alpha \sin x - R \sin \alpha \cos x$
 $= \sin x - \sqrt{3} \cos x$

$\Rightarrow R \cos \alpha = 1$ $\therefore \tan \alpha = \sqrt{3}$, $\alpha = \frac{\pi}{3}$
 $R \sin \alpha = \sqrt{3}$ and $R = \sqrt{1^2 + \sqrt{3}^2} = 2$

(ii) From (i), $2 \sin(x - \frac{\pi}{3}) = \sqrt{2}$
 $\therefore \sin(x - \frac{\pi}{3}) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$



$\therefore x - \frac{\pi}{3} = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$

$\therefore x = \frac{7\pi}{12} \text{ or } \frac{13\pi}{12}$

(c) (i) $u = 4 - x^2$
 $\frac{du}{dx} = -2x$ or $du = -2x dx$ $x=0, u=4$
 $x=\sqrt{3}, u=1$

$\therefore I = -\frac{1}{2} \int_4^1 \frac{du}{\sqrt{u}} = \frac{1}{2} \int_1^4 u^{-\frac{1}{2}} du$
 $= \frac{1}{2} \cdot 2 \left[u^{\frac{1}{2}} \right]_1^4$
 $= 2 - 1 = 1$

(ii) $I = \int_0^{\sqrt{3}} \frac{4}{\sqrt{4-x^2}} - \frac{x}{\sqrt{4-x^2}} dx = 4 \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} - 1$, from (i)
 $= 4 \frac{\pi}{3} - 1$

Q. 3

(a) $f(x) = 2x e^{x^2} - 1$

$\therefore x_1 = 1.2 - \frac{e^{1.44} - 1.2 - 3}{2.4 e^{1.44} - 1} = 1.1977 \dots$

$\therefore$ two decimal approxⁿ = 1.20

(b) (i) $\sin 2A = \frac{2t}{1+t^2}$

(ii) put $t = \tan A$,

then $\operatorname{cosec} 2A - 3 \cot 2A = \frac{1+t^2}{2t} - 3 \cdot \frac{1-t^2}{2t}$

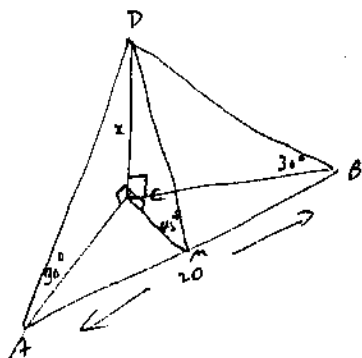
$= \frac{1+t^2 - 3 + 3t^2}{2t}$

$= \frac{4t^2 - 2}{2t}$

$= 2t - \frac{1}{t}$

$= 2 \tan A - \cot A$

(c) (i)

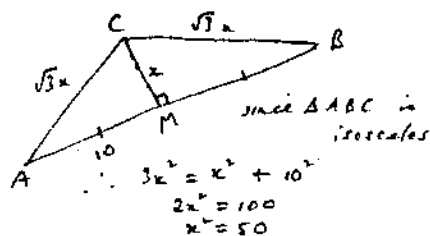


(ii) In $\triangle ACD$,

$\tan 30^\circ = \frac{x}{AC} = \frac{1}{\sqrt{3}}$

$\Rightarrow AC = \sqrt{3}x$

(iii) From (ii), we have

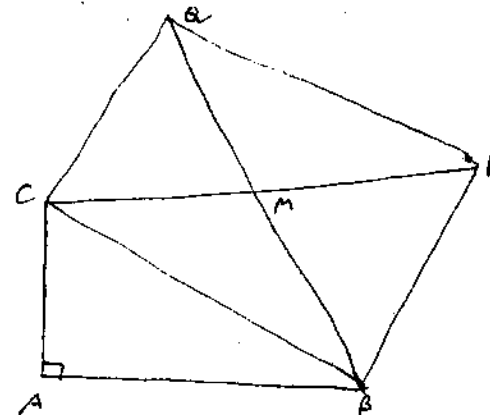


Since $\triangle ABC$ is isosceles
 $3x^2 = x^2 + 10^2$
 $2x^2 = 100$
 $x^2 = 50$

$\therefore x = \sqrt{50} = 5\sqrt{2}$

Q. 4

(i)



(ii) $\angle CMB = 90^\circ$, the diagonals of a square meet at right angles

$\therefore$ In $\square ABMC$, $\angle A + \angle M = 180^\circ$

$\Rightarrow \square ABMC$ is cyclic, opposite angles are supplementary

(iii) $MC = MB$, equal diagonals in a square bisect each other.

$\therefore \angle CAM = \angle BAM$, $\angle$ s at the circumference of a circle standing on equal arcs.

$\therefore MA$ bisects $\angle BAC$

(4) (i) $t = 0, x = 10 \cos 0 = 10$

$\dot{x} = -10 \sin t = -10 \sin 0$ at $t = 0$
 $= 0$

$\therefore$ particle is initially at rest at $x = 10$

(ii) $T = \frac{2\pi}{\omega} \therefore \omega = \frac{2\pi}{T}$

$\therefore b = 10 \cos\left(\frac{2\pi}{T} \cdot \frac{T}{3}\right) = 10 \cos\left(\frac{2\pi}{3}\right)$

i.e. $b = 10\left(-\frac{1}{2}\right) = -5$

$$(iii) \quad \ddot{x} = -10a \sin \omega t = -10 \cdot \frac{2\pi}{T} \sin\left(\frac{2\pi t}{T}\right)$$

$$\therefore -20\sqrt{3} = -\frac{20\pi}{T} \sin\left(\frac{2\pi}{3}\right) = -\frac{20\pi}{T} \cdot \frac{\sqrt{3}}{2}$$

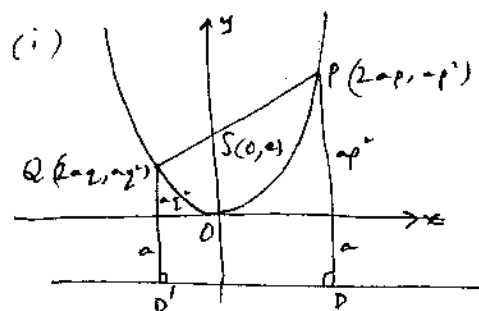
$$\therefore T = \frac{\pi}{2} \text{ (seconds)} \quad \therefore \text{period is } \frac{\pi}{2} \text{ s}$$

$$(C) \quad \binom{n}{3} \div \binom{n-1}{2} = \frac{n!}{(n-3)! 3!} \div \frac{(n-1)!}{(n-3)! 2!}$$

$$= \frac{n!}{(n-3)! 3!} \times \frac{(n-3)! 2!}{(n-1)!} = \frac{n}{3}$$

Q 5

(a) (i)



$$\begin{aligned} \text{gradient } PQ &= \frac{ap^2 - az^2}{2ap - 2az} \\ &= \frac{-p^2 + z^2}{2a(p-z)} \\ &= \frac{-(p-z)(p+z)}{2a(p-z)} \\ &= \frac{p+z}{2} \end{aligned}$$

$$\therefore \text{chord } PQ \text{ is } y - ap^2 = \frac{1}{2}(p+z)(x - 2ap)$$

$$\text{or } y - ap^2 = \frac{1}{2}(p+z)x - ap(p+z)$$

$$\Rightarrow y - \frac{1}{2}(p+z)x + apz = 0$$

(ii) Since $S(0, a)$ is on PQ , then

$$a - 0 + apz = 0$$

$$\therefore pz = -1 \quad \text{or } z = -\frac{1}{p}$$

(iii) $PQ = PS + SQ = PD + QD'$, focus-directrix defn (see diagram)

$$= a + ap^2 + a + az^2, \quad z = -\frac{1}{p}$$

$$= 2a + a\left(p^2 + \frac{1}{p^2}\right)$$

(iv) If PQ is a diameter, the radius is $a + \frac{a}{2}\left(p^2 + \frac{1}{p^2}\right)$,
from (iii)

$$\therefore \text{the centre is } \left(\frac{2ap + 2az}{2}, \frac{ap^2 + az^2}{2}\right)$$

$$= \left(a\left(p - \frac{1}{p}\right), \frac{a}{2}\left(p^2 + \frac{1}{p^2}\right)\right), \quad z = -\frac{1}{p}$$

$\therefore$ distance from centre to directrix $y = -a$ is

$$\frac{a}{2}\left(p^2 + \frac{1}{p^2}\right) + a = \text{radius}$$

$\therefore$ directrix is a tangent to the circle

(b) (i) $A = 6 \times 10^2 + 12.6t = 600 + 12.6t$

(ii) If an edge is x , $A = 6x^2$, $V = x^3$

$\therefore$ From (i), $6x^2 = 600 + 12.6t$

$x^2 = 100 + 2.1t$

$\therefore V = (100 + 2.1t)^{3/2}$

$\therefore$ So, $\frac{dV}{dt} = \frac{3}{2} (100 + 2.1t)^{1/2} (2.1)$

$= 3.15 \times \sqrt{121} \text{ cm}^3/\text{s}$ when $t=10$

$= 34.65 \text{ cm}^3/\text{s}$

Alternatively, using $A = 6x^2$, $V = x^3$, $\frac{dA}{dt} = 12.6$,

we have $\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$

$= \frac{dV}{dx} \cdot \frac{dx}{dA} \cdot \frac{dA}{dt}$

$= 3x^2 \cdot \frac{1}{12x} \cdot (12.6)$

$= 3.15x$

But, when $t=10$, $A = 6 \times 10^2 + 12.6 \times 10 = 726$

$\therefore 6x^2 = 726 \Rightarrow x = 11$

$\therefore \frac{dV}{dt} = 3.15 \times 11 \text{ cm}^3/\text{s} = 34.65 \text{ cm}^3/\text{s}$

Qn 6

(a) $E(0) = 9^2 - 4^0 = 80$ is a multiple of 5

$\therefore$ assume $E(n) = 9^{n+2} - 4^n = 5q$, q an integer, $n \geq 0$

Then, $E(n+1) = 9^{n+3} - 4^{n+1}$

$= 9(9^{n+2}) - 4^{n+1}$

$= 9(5q + 4^n) - 4^{n+1}$, using the assumption

$= 5(9q) + 4^n(9-4)$

$= 5(9q + 4^n)$ is a multiple of 5

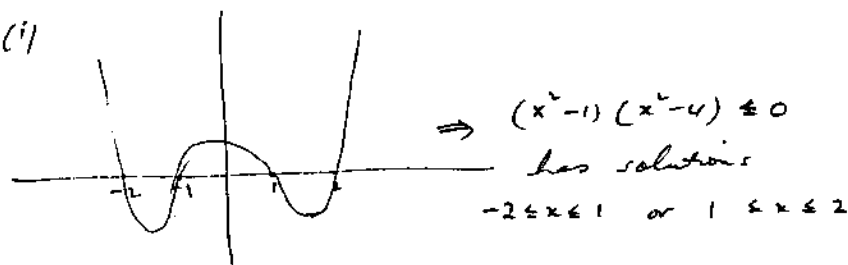
since $9q + 4^n$ is an integer

$\therefore$ if $E(n)$ is a multiple of 5, so is $E(n+1)$

But, $E(0)$ is a multiple of 5

$\therefore E(n)$ is a multiple of 5 by induction

(b) (i)



(ii) (L) $\frac{d(\frac{1}{2}v^2)}{dx} = 10x - 4x^3$

$\therefore \frac{1}{2}v^2 = 5x^2 - x^4 + C$, C a constant

i.e. $\frac{1}{2}v^2 + x^4 - 5x^2 = C$

When $x = \sqrt{2}$, $v = 2$

$\therefore 2 + 4 - 10 = C = -4$

(B) we have $\frac{1}{2}v^2 = 5x^2 - x^4 - 4$

$$\text{or } \frac{1}{2}v^2 = -(x^4 - 5x^2 + 4)$$

$$= -(x^2 - 1)(x^2 - 4)$$

$$\text{Now, } \frac{1}{2}v^2 \geq 0 \Rightarrow (x^2 - 1)(x^2 - 4) \leq 0$$

Using (b)(i) and when $x = \sqrt{2}$, $v = 2$, we have the particle oscillates between $x = 1$ and $x = 2$

(C) (i) $P(5 \text{ males, } 5 \text{ females}) = \binom{10}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^5$
 $= 0.246$, 3 d.p.

(ii) $P(\text{more females}) = P(\text{more males})$, since $P(M) = P(F) = \frac{1}{2}$
 $\therefore$ using (i), $P(\text{more females}) = \frac{1 - 0.246}{2} = 0.377$

Qn 7

(a) if $x+1 > 0$, then $-2x > 0$

$$\text{i.e. } x > -1$$

$$\text{i.e. } x < 0$$

$\therefore$ solution is $-1 < x < 0$

if $x < -1$, we'd have $x > 0 \Rightarrow$ no further solutions

$\therefore -1 < x < 0$

(b)(i) We need $\frac{-2x}{x+1} > 0$ and $-2x > 0$ and $x+1 > 0$.

All 3 inequalities are "true" from (a)

(ii) From (i), $y = \ln(2x) - \ln(x+1)$

$$\therefore \frac{dy}{dx} = \frac{-2}{-2x} - \frac{1}{x+1}$$

$$= \frac{1}{x} - \frac{1}{x+1} = \frac{x+1-x}{x(x+1)}$$

$$= \frac{1}{x(x+1)} \neq 0 \text{ for any } x$$

$\therefore$ there are no stationary points

(iii) Since $-1 < x < 0$, then $x(x+1) < 0$

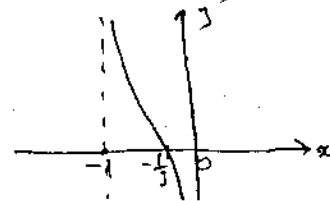
$$\therefore \text{From (ii)} \frac{dy}{dx} < 0 \text{ for } -1 < x < 0$$

i.e. curve is decreasing for $-1 < x < 0$

When $y = 0$, $\frac{-2x}{x+1} = 1$

$$\therefore -2x = x+1$$

$$\Rightarrow x \text{ intercept is } -\frac{1}{3}$$



(iv) Since curve is decreasing, the inverse function

$$\text{is } x = \ln\left(\frac{-2y}{y+1}\right)$$

$$\therefore \frac{-2y}{y+1} = e^x$$

$$-2y = e^x y + e^x$$

$$\therefore y(e^x + 2) = -e^x$$

$$\therefore \text{inverse function is } y = -\frac{e^x}{e^x + 2}$$

$$(v) A = \left| \int_0^2 x \, dy \right| = \int_0^2 \frac{e^y}{e^y + 2} \, dy, \text{ from (iv)}$$

$$= \left[\ln(e^y + 2) \right]_0^2$$

$$= \ln(e^2 + 2) - \ln 3 \quad \text{u}^2$$